

ROLANDO S. GARCIA, PH.D.

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EDUCATION

University of California, Berkeley	Berkeley, CA
Ph.D. in Computer Science advised by Dr. Joseph M. Hellerstein	2017 – 2024
Dissertation: <i>The Management of Context in the Machine Learning Lifecycle</i>	
Arizona State University	Tempe, AZ
B.S. in Computer Science, <i>Summa Cum Laude</i>	2014 – 2017
Northern Arizona University	Flagstaff, AZ
Philosophy and Computer Information Systems	2010 – 2013

AWARDS AND FELLOWSHIPS

ASU Presidential Postdoctoral Fellowship, two years of funding.	2024
Dell Technologies Inc. Research Award, one year of funding.	2023
NSF Graduate Research Fellowship, three years of funding.	2017
UC Berkeley Chancellor's Fellowship, two years of funding.	2017
UC Berkeley's EECS Excellence Award.	2017
CRA Outstanding Undergraduate Researcher Award, honorable mention.	2017
Fulton Undergraduate Research Initiative, research award.	2016
IEEE VAST Grand Challenge Award, outstanding comprehensive submission.	2015

RESEARCH

School of Computing & AI scai.engineering.asu.edu	July 2024 – Present
Presidential Postdoctoral Fellow, advised by Dr. Ross Maciejewski	<i>Arizona State University</i>

- *Submitted and under review:*
 - * NSF Future CoRE: *Interactive Graph Queries: Explainable GNN Approaches to Ad-Hoc Queries on Heterogeneous Networks*, w/ Yixuan He¹ et al.
 - Co-PI at 20% effort, \$1M over 3 years.
 - * NSF TTP-T: *Scalable Human-in-the-Loop Review Systems*, w/ R. Maciejewski, & T. Winarski²
- *In progress:*
 - * ReviewerApp: Visual Co-Display of LLM and Data Mining Signals, w/ Fan Lei et al.
 - Mentoring ASU undergraduate students in computer science (B.S.) Akshit Jain, Rohan Boddu.
 - * Field Study: Logging & Record-Keeping Practices in Robotics Research, w/ Erick Duarte et al.

RISE Lab rise.cs.berkeley.edu	June 2017 – June 2024
Graduate Student Researcher & Fellow, advised by Dr. Joe Hellerstein	<i>University of California, Berkeley</i>
• FlorDB: a hindsight logging database for the machine learning lifecycle. • ... How Engineers Operationalize Machine Learning, w/ Shreya Shankar et al. • Internships: Linea Labs (<i>startup</i>).	github.com/ucbrise/flor

Amazon A9 (Search)	May 2018 – December 2018
Research Intern, Search Infrastructure Team	<i>Palo Alto, CA</i>
• Studied end-to-end ML pipelines orchestrated via AWS SageMaker, spanning ingestion, transformation, tokenization, training, and evaluation across multi-team workflows. • Observed and contributed to DAG-based pipeline design for large-scale search model training and A/B testing.	

¹School of Mathematical and Natural Sciences

²Sandra Day O'Connor College of Law

PEER-REVIEWED PUBLICATIONS

- Flow with FlorDB: Incremental Context Maintenance for the ML Lifecycle** CIDR 2025
Rolando Garcia, Pragma Kallanagoudar, Chithra Anand, Sarah E. Chasins,
Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.
Conference on Innovative Data Systems Research, 2025.
- “We Have No Idea How Models will Behave in Production until Production”:
How Engineers Operationalize Machine Learning** CSCW 2024
Shreya Shankar*, *Rolando Garcia**, Joseph M. Hellerstein, Aditya G. Parameswaran.
ACM Conf’ on Computer Supported Cooperative Work and Social Computing 2024.
*Co-first authors. **Honorable Mention:** MLOps Community Best Paper Award
- Quilt: Custom UIs for Linking Unstructured Documents
to Structured Dataasets** UIST Demo 2024
Pragma Kallanagoudar, Chithra Anand, *Rolando Garcia*, Rebecca M. M. Hicke,
Aditya G. Parameswaran, Eunice Jun, Sarah E. Chasins. *UIST Adjunct ’24:
Proceedings of the 37th Annual ACM Symposium on User Interface Software
and Technology*, 2024.
- Hindsight Logging for Model Training** VLDB 2021
Rolando Garcia, Eric Liu, Vikram Sreekanti, Bobby Yan, Anusha Dandamudi,
Joseph E. Gonzalez, Joseph M. Hellerstein, Koushik Sen. *Proceedings of the
VLDB Endowment*, 14(4): 682-693, 2021.
- Context: The Missing Piece in the Machine Learning Lifecycle** KDD CMI 2018
Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw,
Joseph E. Gonzalez, Joseph M. Hellerstein.
Workshop on Common Model Infrastructure at KDD, 2018.
- A Visual Analytics Framework for Exploring Theme Park Dynamics** ACM TIIS 2018
Michael Steptoe, Robert Krüger, *Rolando Garcia*, Xing Liang, Ross Maciejewski.
ACM Transactions on Interactive Intelligent Systems, 2018.
- The State-of-the-Art in Predictive Visual Analytics** CGF 2017
Yafeng Lu, *Rolando Garcia*, Brett Hansen, Michael Gleicher, Ross Maciejewski.
Computer Graphics Forum, 2017.
- VADER/VIS Grand Challenge Award** IEEE VAST 2015
Michael Steptoe, Robert Krüger, Yifan Zhang, Xing Liang, *Rolando Garcia*,
Sagarika Kadambi, Wei Luo, Thomas Ertl, Ross Maciejewski.
Proceedings of the IEEE VAST Challenge Workshop, 2015.

TALKS

- Incremental Context Maintenance for the ML Lifecycle** CIDR 2025
Amsterdam
Rolando Garcia, Pragma Kallanagoudar, Chithra Anand, Sarah E. Chasins,
Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein.
Conference on Innovative Data Systems Research. January 2025.

How Engineers Operationalize Machine Learning Shreya Shankar, <i>Rolando Garcia</i> , Joseph M. Hellerstein, Aditya G. Parameswaran. <i>ACM Conf' on Computer Supported Cooperative Work and Social Computing</i> . November 2024.	CSCW 2024 Costa Rica
Flow with FlorDB: Agile MLOps without Compromises <i>Rolando Garcia</i> . Colorado School of Mines. October 2024.	Mines Seminar, 2024 Golden, CO
Operationalizing Machine Learning: An Interview Study <i>Rolando Garcia</i> , Shreya Shankar. December 2022.	Flyte, 2022
Operationalizing Machine Learning: An Interview Study Shreya Shankar, <i>Rolando Garcia</i> . Berkeley, CA, October 2022.	Inaugural Address, 2022 EPIC, UC Berkeley
Operationalizing Machine Learning: An Interview Study Shreya Shankar, <i>Rolando Garcia</i> . October 2022.	Canva, 2022
Operationalizing Machine Learning: An Interview Study Shreya Shankar, <i>Rolando Garcia</i> . San Francisco Bay Area, October 2022.	MLOps World, 2022 San Francisco
FLOR: Managing the Infinite Virtual Metadata of Model Training <i>Rolando Garcia</i> . Cornell University, February 2022.	Cornell Seminar, 2022
Hindsight Logging for Model Training <i>Rolando Garcia</i> , Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan, Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. <i>International Conference</i> <i>on Very Large Databases</i> . August 2021.	VLDB 2021 Copenhagen
Choose What to Log, After you Execute <i>Rolando Garcia</i> , Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan, Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. Retreat talk to lab sponsors. Monterey, CA. January 2019.	RISE Retreat 2019
ML in Context: Preserve, Share, and Analyze Experiments with Flor <i>Rolando Garcia</i> , Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein. Talk and Tutorial for lab sponsors. Berkeley, CA. October 2018.	RISE Camp 2018
Context: The Missing Piece in the Machine Learning Lifecycle <i>Rolando Garcia</i> , Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Joseph Gonzalez, Joseph Hellerstein. <i>The 24th ACM SIGKDD Intl' Conf</i> . August 2018.	KDD 2018 London
Data Management Challenges in the Machine Learning Lifecycle <i>Rolando Garcia</i> , Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein. Invited talk at Amazon Search. Palo Alto, CA. April 2018.	Amazon 2018 Palo Alto
Design, Run, and Reproduce Experiments with Jarvis <i>Rolando Garcia</i> , Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein. Retreat talk to lab sponsors. Monterey, CA. January 2018.	RISE Retreat 2018

Jarvis: Managing the Machine Learning Lifecycle

Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein.
Invited talk at Google. Mountain View, CA. September 2017.

Google 2017
Mountain View

The State-of-the-Art in Predictive Visual Analytics

Yafeng Lu, *Rolando Garcia*, Brett Hansen, Michael Gleicher, Ross Maciejewski.
19th *EG/VGTC Conference on Visualization*, Barcelona, Spain. June 2017.

EuroVis 2017
Barcelona

Automatic Design and Optimization of Deep Learning Models

Rolando Garcia, Randal Olson, Jason Moore.
The Leadership Alliance National Symposium, Stamford, CT. July 2016.

LANS 2016

VAST 2015 Grand Challenge – Team VADER/VIS

Michael Steptoe, Robert Krüger, Yifan Zhang, Xing Liang, *Rolando Garcia*, Sagarika Kadambi, Wei Luo, Thomas Ertl, Ross Maciejewski.
IEEE Visual Analytics Science and Technology Conference, Chicago, IL. October 2015.

VAST 2015
Chicago

TEACHING**Introduction to Database Systems (COMPSCI 186/286)**

Graduate Student Instructor, 450 students

Spring 2018
University of California, Berkeley

Introduction to Programming Languages (CSE 240)

Undergraduate Student Instructor, 267 students

Fall 2015
Arizona State University, Tempe

Introduction to Ethics (PHI 105)

Undergraduate Student Instructor, 50 students

Fall 2013
Northern Arizona University, Flagstaff

SERVICE**Diversifying Access to Research in Engineering (DARE)**

Organizer | dare.berkeley.edu

June 2020 – December 2022
University of California, Berkeley

EECS Graduate Admissions

Graduate Student Representative | eecs.berkeley.edu

Fall 2021
University of California, Berkeley

Data Systems and Foundations Seminar

Coordinator | dsf.berkeley.edu

Fall 2020
University of California, Berkeley